5.2 THE LS TECHNIQUE

The principle of LS is used in this chapter for signals of one variable, although the concept applies equally well for signals with more than one variable. Furthermore, we study discrete signals, which can always be derived from continuous signals if the sampling frequency is sufficient high enough so that aliasing does not take place.

Let us, for example, have a discrete signal that is represented in its vector form as follows: $\boldsymbol{x} = [x(1)\ x(2)\ \ldots\ x(N)]$. We would like, next, to approximate the function with a polynomial of the form:

$$\widehat{x}(nT) = c_1 + c_2 \times (nT) + c_3 \times (nT)^2 + \ldots + c_M \times (nT)^{M-1} \quad T = \text{sampling time} \tag{5.1}$$

Since the approximating polynomial is a linear function of the unknown c_i's, we are dealing with **linear LS approximation**. The difference between the exact and the approximate function is the error. We can define a **cost function** $\boldsymbol{J}$, which is equal to the sum of the difference of errors squared, the total square error. Hence, we write

$$J = \sum_{n=1}^{N} [x(n) - \widehat{x}(n)]^2 \tag{5.2}$$

Example 5.2.1

Let $x(n)$ =2sin(0.05πn)+randn(1,20), with $T = 1$, be the signal to be approximated. Let the signal $\widehat{x}(nT) = c_1 + c_2 n + c_3 n^2$, the sampling time $T = 1$, approximate the original signal. Find the unknown constants c_i's and plot the results.

Solution: The cost function for this case is

$$J = \sum_{n-1}^{20} [x(n) - \widehat{x}(n)]^2 = \sum_{n=1}^{20} [x(n) - c_1 - c_2 n - c_3 n^2]^2$$

The partial derivatives with respect to c_i' s are

$$\boxed{\begin{aligned}
\frac{\partial J}{\partial c_1} &= -2\sum_{n=1}^{20}[x(n) - c_1 - c_2 n - c_3 n^2] = 0 \ \text{ or} \\
&\sum_{n=1}^{20} x(n) - c_1 \sum_{n=1}^{20} 1 - c_2 \sum_{n=1}^{20} n - c_3 \sum_{n=1}^{20} n^2 = 0 \\
\frac{\partial J}{\partial c_2} &= -2\sum_{n=1}^{20}[x(n) - c_1 - c_2 n - c_3 n^2]\, n = 0 \ \text{ or} \\
&\sum_{n=1}^{20} x(n)n - c_1 \sum_{n=1}^{20} n - c_2 \sum_{n=1}^{20} n^2 - c_3 \sum_{n=1}^{20} n^3 = 0 \\
\frac{\partial J}{\partial c_3} &= -2\sum_{n=1}^{20}[x(n) - c_1 - c_2 n - c_3 n^2] n^2 = 0 \ \text{ or} \\
&\sum_{n=1}^{20} x(n)n^2 - c_1 \sum_{n=1}^{20} n^2 - c_2 \sum_{n=1}^{20} n^3 - c_3 \sum_{n=1}^{20} n^4 = 0
\end{aligned}}$$

Using MATLAB, we obtain the following system:

$$\begin{aligned}
20c_1 + 210c_2 + 2870c_3 &= 27.7347 \\
210c_1 + 2870c^2 + 44100c_3 &= 296.1023 \\
2870c_1 + 44100c_2 + 722666c_3 &= 3499.9838
\end{aligned}$$

The values of the unknown c_i's are found by using the expression:

The values of the unknown c_i's are found by using the expression:

$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 20 & 210 & 2370 \\ 210 & 2870 & 44100 \\ 2870 & 44100 & 722666 \end{bmatrix}^{-1} \begin{bmatrix} 27.7347 \\ 296.1023 \\ 3499.9838 \end{bmatrix} = \begin{bmatrix} -1.2457 \\ 0.7042 \\ -0.0332 \end{bmatrix}$$

Therefore, the estimate curve is given by

$$\widehat{x}(n) = -1.2457 + 0.7042n - 0.0332n^2$$
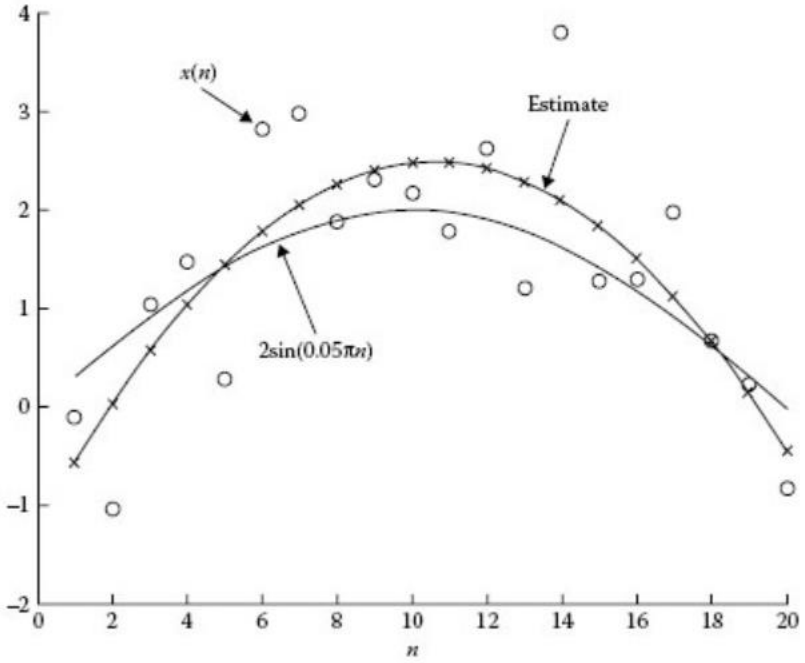
Figure 5.1 shows the original signal $\{x(n)\}$, the approximate one $\{\widehat{x}(n)\}$, based on the LS approach, and the undisturbed signal $\{2\sin(0.05)\pi n\}$.